

JavaScript Beginner Notes

Easy handwritten-notes style study material

1. Introduction to JavaScript

JavaScript is a high-level programming language used to make web pages interactive and dynamic.

- Lightweight and easy to learn.
- Runs in web browsers.
- Supports object-oriented programming.
- Used for frontend and backend development.
- Handles events, animations, and user input.

2. Adding JavaScript to HTML

Internal JavaScript is written inside the `<script>` tag in an HTML file.

```
<!DOCTYPE html>
<html>
<body>
  <script>
    console.log("Hello World");
  </script>
</body>
</html>
```

External JavaScript

External JavaScript is stored in a separate .js file and connected to HTML.

```
<script src="script.js"></script>

// script.js
console.log("Hello World");
```

3. Variables

A variable is a named storage location used to store data.

- var: function-scoped; can be redeclared.
- let: block-scoped; can be reassigned.
- const: block-scoped; cannot be reassigned.

Variable Example

```
let age = 20;
const pi = 3.14;
var name = "Shashank";

console.log(age);
console.log(pi);
console.log(name);
```

4. Data Types

A data type specifies the kind of value stored in a variable.

- Number
- String
- Boolean
- Undefined
- Null
- BigInt
- Symbol

Data Type Example

```
let age = 20;           // Number
let name = "Shashank"; // String
let isStudent = true;  // Boolean
let x;                 // Undefined
let value = null;      // Null
```

5. Operators

Operators are symbols used to perform operations on values.

- Arithmetic: +, -, *, /, %, **
- Assignment: =, +=, -=, *=
- Comparison: ==, ===, !=, !==, >, <
- Logical: &&, ||, !

Operator Example

```
let a = 10;
let b = 3;

console.log(a + b); // 13
console.log(a % b); // 1
console.log(a > b); // true
```

6. Conditional Statements

Conditional statements execute code based on a condition.

```
let age = 20;

if (age >= 18) {
  console.log("Eligible to vote");
} else {
  console.log("Not eligible to vote");
}

let marks = 75;

if (marks >= 90) {
  console.log("A Grade");
} else if (marks >= 60) {
  console.log("B Grade");
} else {
  console.log("C Grade");
}
```

7. Loops

Loops execute a block of code repeatedly.

```
for (let i = 1; i <= 5; i++) {  
  console.log(i);  
}  
  
let i = 1;  
  
while (i <= 5) {  
  console.log(i);  
  i++;  
}
```

8. Functions

A function is a reusable block of code that performs a particular task.

```
function add(a, b) {  
  return a + b;  
}  
  
console.log(add(5, 3)); // 8  
  
const square = (n) => n * n;  
  
console.log(square(4)); // 16
```

Quick Revision

- JavaScript makes websites interactive.
- Use let and const for modern JavaScript.
- Variables store values.
- Operators perform calculations and comparisons.
- Conditions make decisions.
- Loops repeat code.
- Functions make code reusable.